EECS 16A Designing Information Devices and Systems I
Discussion 14A

1. Search and Rescue Dogs

Berkeley's Puppy Pound needs your help! While Mr. Muffin was being walked, the volunteer let go of his leash and he is now running wild in the streets of Berkeley (which are quite dangerous)! Thankfully, all of the puppies at the pound have a collar that sends a bluetooth signal to receiver towers, which are spread throughout the streets (pictured below). If the puppy/collar is within range of the receiver tower, the collar will send the tower a message: the distance of the collar to the tower. Each cell tower has a range of 5 city blocks. Can you help the pound locate their lost puppy?

Note: A city block is defined as the middle of an intersection to the middle of an adjacent intersection (scale provided on map.) Mr. Muffin is constrained to running wild in the streets, meaning he won't be found in any buildings. If your TA asks 'Where is Mr. Muffin?' it is sufficient to answer with 'this intersection' or 'between these two intersections.'

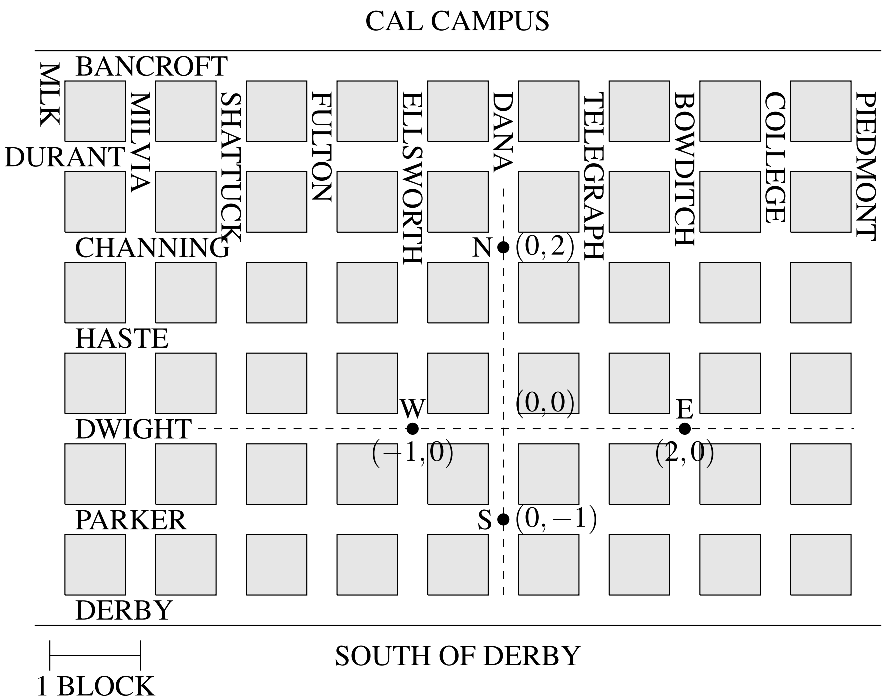

[1]

(a) You check the logs of the cell towers, and they have received the following messages:

Sensor	Distance
N	5
W	$\sqrt{20}$
E	$\sqrt{5}$
S	$\sqrt{10}$

On the map provided, identify where Mr. Muffin is!

[1] http://www.pupsmile.com/wp-content/uploads/2012/11/running_happy_dog-1024x684.jpeg


CAL CAMPUS
BANCROFT
DURANT
CHANNING
HASTE
DWIGHT
PARKER
DERBY
MLK
MILVIA
SHATTUCK
FULTON
ELLSWORTH
DANA
TELEGRAPH
BOWDITCH
COLLEGE
PIEDMONT
N (0,2)
(0,0)
W (−1,0)
E (2,0)
S (0,−1)
SOUTH OF DERBY
1 BLOCK

(b) Can you set this up as a system of equations? Is it linear? If it's not linear, can you think of a way to make it linear? Now, how do you set this up in matrix form?

Hint: Set $(0,0)$ to be Dwight and Dana.

Hint 2: Distance $= \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$.

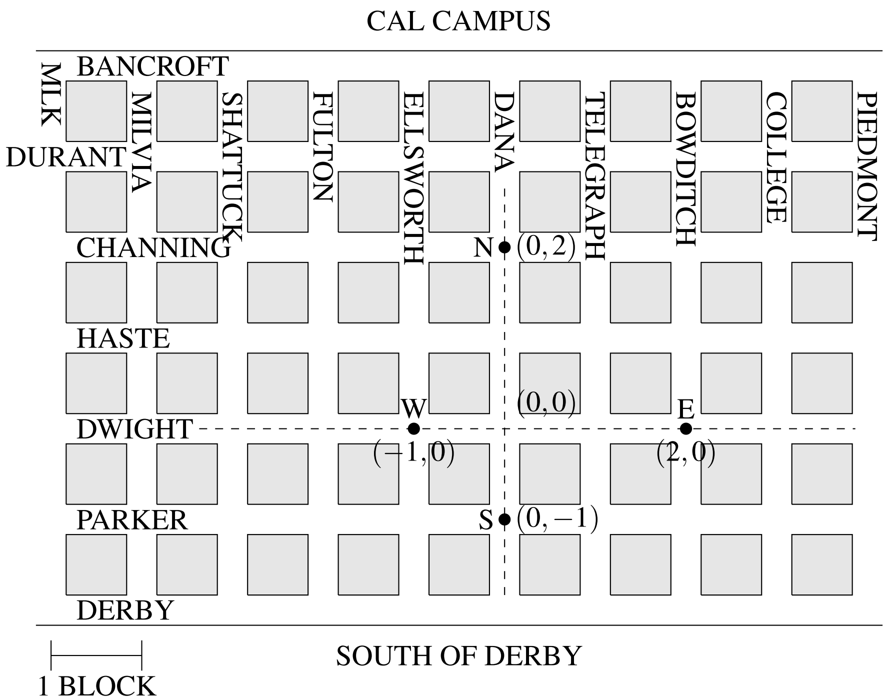
Hint 3: You don't need all 4 equations. You have two unknowns, x and y. You know from lecture that you need three circles to uniquely find a point. How can you use the third circle/equation to get two equations and two unknowns?

Note: Remember to check for consistency for all nonlinear equations after finding the coordinates.

(c) Suppose Mr. Muffin is moving fast, and by the time you get to destination in part (a) he's already run off! You check the logs of the cell towers again, and see the following updated messages:

Sensor	Distance
N	5
W	$\sqrt{20}$
E	Out of Range
S	Out of Range

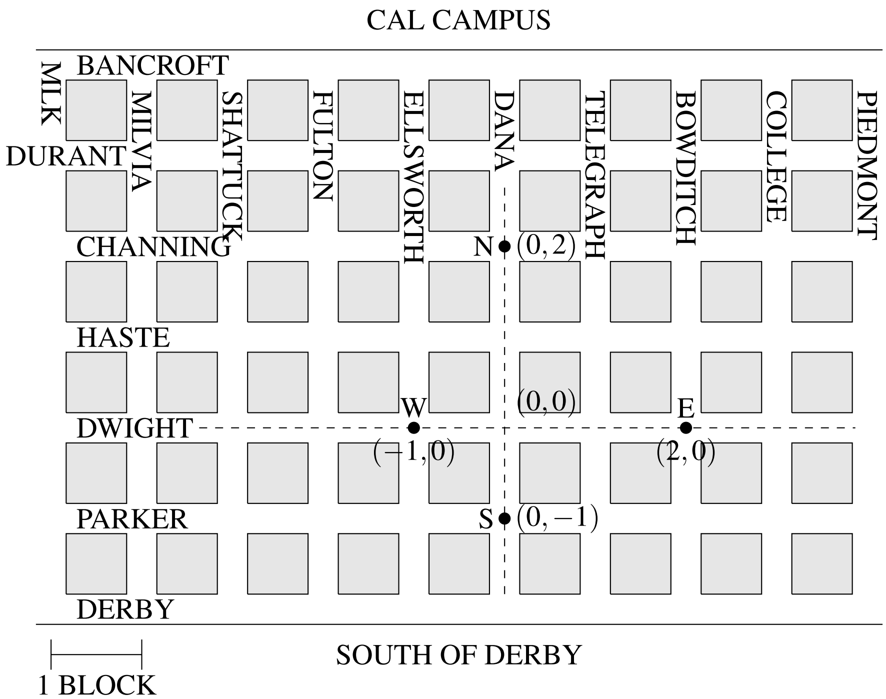
Can you find Mr. Muffin? (*Note:* Each cell tower has a range of 5 city blocks.)



(d) Mr. Muffin is a very mischievous puppy, and while playing and running around he damaged his collar. The transmitter on his collar will still send a signal to the receiver towers, but the distance sensor has noise. You check the logs of the cell towers, and they have received the following messages:

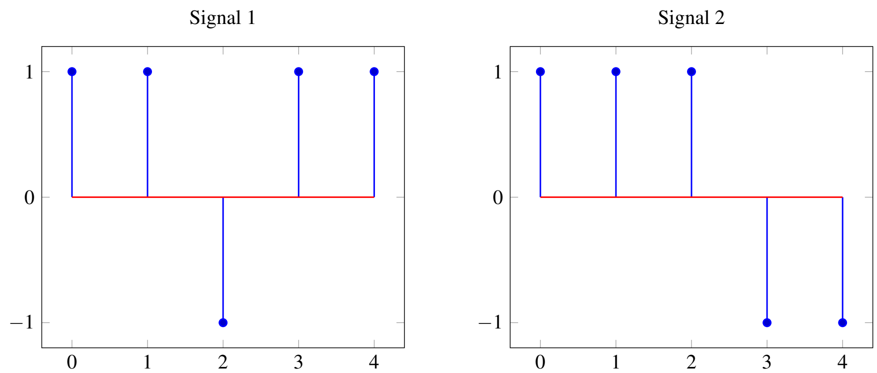
Sensor	Distance
N	4.5 ± 0.5
W	Out of Range
E	4.5 ± 0.5
S	Out of Range

On the map provided, identify where Mr. Muffin is! Can you find exactly where he is? (*Note:* Each cell tower has a range of 5 city blocks.)

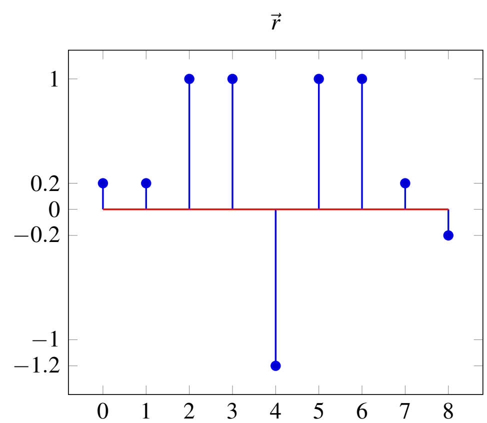


2. Identifying satelites and their delays

We are given the following two signals, $\vec{s}_1$ and $\vec{s}_2$ respectively, that are signatures for two satellites.

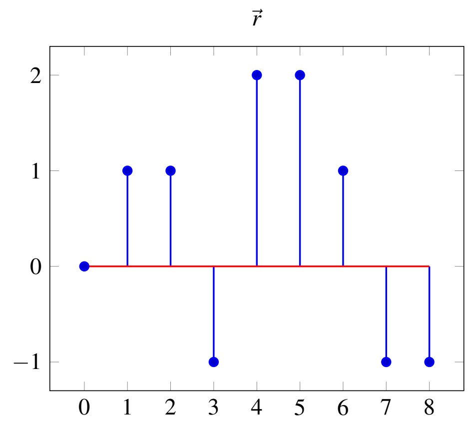


(a) Your cellphone antenna receives the following signal $r[n]$. You know that there may be some noise present in $r[n]$ in addition to the transmission from the satellite.



Which satellites are transmitting? What is the delay between the satellite and your cellphone? Use cross-correlation to justify your answer. You can use iPython to compute the cross-correlation.

(b) Now your cellphone receives a new signal $r[n]$ as below. Which satellites are transmitting and what is the delay between each satellite and your cellphone?


$\vec{r}$
2
1
0
−1
0
1
2
3
4
5
6
7
8

3. From Inner Products To Projections

Given that $\langle \vec{x}, \vec{y} \rangle$ is a measure of similarity between two vectors, let's try to use this to find how much of one vector $\vec{y}$ is in the direction of another vector $\vec{x}$.

(a) Let's start with $\langle \vec{x}, \vec{y} \rangle$. We want a quantity that is independent of the norm of $\vec{x}$, $\|\vec{x}\|$. Is $\langle \vec{x}, \vec{y} \rangle$ independent of the norm? Consider $\langle \vec{x}, \vec{y} \rangle$ for the examples below.
$\vec{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

(b) Suppose we divide $\langle \vec{x}, \vec{y} \rangle$ by the norm of $\vec{x}$, $\|\vec{x}\|$, to get $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|}$. Is this new quantity independent of the norm of $\vec{x}$? Test it on the examples above.

(c) We now have a scalar quantity that represents how much of $\vec{y}$ is in the direction of $\vec{x}$. Now let's look for a vector $\vec{z}$ that has a norm of $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|}$ and points in the same direction as $\vec{x}$. The vector $\vec{z}$ should be written in terms of $\vec{x}$ and $\vec{y}$.

(d) Consider the quantity $\frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|\|\vec{y}\|}$. What is the maximum this quantity could be? When does this occur? What is the minimum this quantity could be? When does this occur?
Hint: Use the Cauchy-Schwartz inequality.

(e) We define the angle between two vectors as $\cos(\theta) = \frac{\langle \vec{x}, \vec{y} \rangle}{\|\vec{x}\|\|\vec{y}\|}$. When do two vectors have an angle of 90° between them? When do they have an angle of 0°? When do they have an angle of 180°?

